

GOT-Computer-Vision

October 1, 2026

#THE DEFENSE OF WESTEROS

The winter is coming, and the Night's Watch is preparing their defense against the more than possible attack from the White Walkers, a one hundred thousand man army that lurks in the darkness. To prevent it, Lord Commander has entrusted us with analyzing images that our ravens get from beyond the wall. The goal is to analyze these images to find and detect enemies and possible threats by implementing histograms, look-up-tables and different techniques for image processing that we are going to explain further in this notebook.

```
[1]: try:
      from google.colab import drive
      drive.mount('/content/drive')
      # If your notebook and images are in your Drive, change directory
      ↪ accordingly:
      %cd '/content/drive/Othercomputers/Mi portátil/cuadernos_extra'
except ImportError:
    pass

import numpy as np
from scipy import signal
import cv2
import matplotlib.pyplot as plt
import matplotlib
from ipywidgets import interactive, fixed, widgets
matplotlib.rcParams['figure.figsize'] = (15.0, 15.0)
import random

images_path = './images/'
```

Mounted at /content/drive

/content/drive/Othercomputers/Mi portátil/cuadernos_extra

###

CHAPTER I: THE NIGHT AT WINTERFELL

For this first part, we are going to inspect the images from our ravens and try to improve the global contrast in order to make our job easier. This will involve analyzing the pixel distribution through histograms to understand the current state of the image and then applying techniques to spread out these pixel intensities, enhancing visibility of critical details that might otherwise be hidden in the low-contrast blizzard conditions.

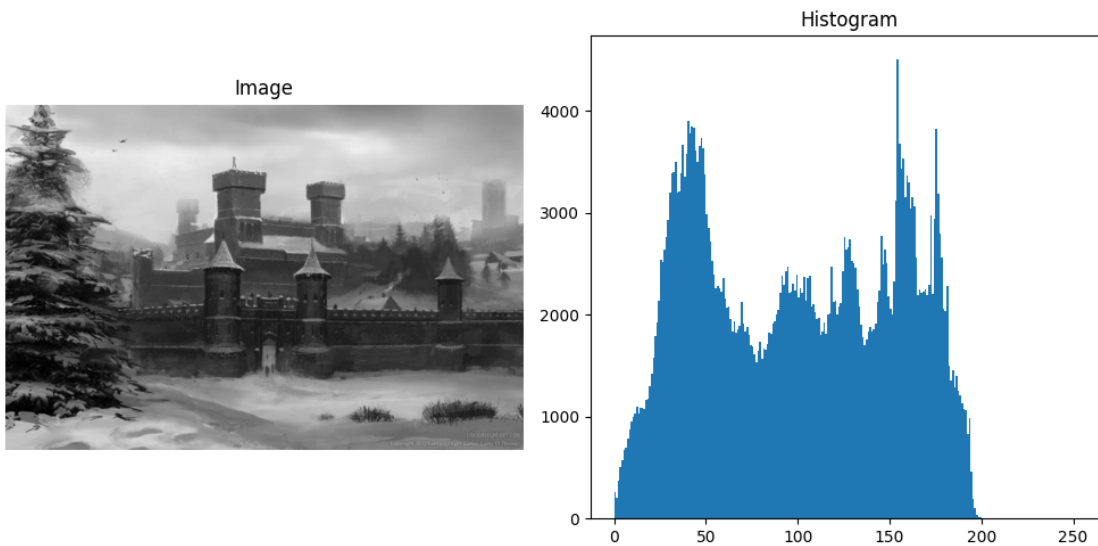
```
[2]: image = cv2.imread(images_path + 'winterfell_blizzard.jpg', cv2.
    ↪IMREAD_GRAYSCALE)

plt.figure(figsize=(10, 5)) # Increased figure size
plt.subplots_adjust(wspace=0.3) # Add spacing between subplots

plt.subplot(1,2,1)
plt.title("Image")
plt.imshow(image, cmap='gray')
plt.axis('off') # Hide axes for cleaner image display

# Now, show its histogram
plt.subplot(1,2,2)
plt.title("Histogram")
plt.hist(image.ravel(), 256, range=[0, 256])

plt.tight_layout() # Adjust layout to prevent overlapping
plt.show()
```



By seeing the histogram of our picture, we can determine that the image has a big concentration of pixels between 0 and 200, but there are no pixels higher than 200. This indicates a low-contrast image, especially in the brighter regions, where details might be lost due to the compressed range of pixel intensities. To effectively identify threats and features in these challenging conditions, we need to enhance the contrast, which histogram equalization can help achieve by distributing the pixel values across the full dynamic range.

0.0.1 HISTOGRAM EQUALIZATION

After the night watch of the ravens, they bring us completely dark images, where, in addition, the snow and fog make it difficult for us to detect enemies. For this, we are going to use histogram equalization to differentiate them. As we can see, in dark images, all pixels are concentrated in the range [0,30] of our histogram, making it impossible to see anything in the dark. With histogram equalization, we will make the pixel intensities more evenly distributed across the histogram, and thus be able to distinguish the silhouettes of the white walkers.

```
[3]: import cv2
import matplotlib.pyplot as plt
import numpy as np

def equalize_chart(image, verbose=False):
    """ Applies histogram equalization to the luminance channel of an image in
    ↪ YCrCb space.

    Args:
        image (numpy.ndarray): Input image in BGR color space.
        verbose (bool): If True, plots the original and processed images
    ↪ along with their histograms.

    Returns:
        numpy.ndarray: Equalized image in BGR color space.
    """
    # 1. Transformación al espacio de color YCrCb desde BGR
    ycrb_image = cv2.cvtColor(image, cv2.COLOR_BGR2YCrCb)

    # 2. Ecualización del canal de luminancia (Y, canal 0)
    ycrb_image[:, :, 0] = cv2.equalizeHist(ycrb_image[:, :, 0])

    # 3. Reconversión al espacio de color BGR original para el retorno
    out_image = cv2.cvtColor(ycrb_image, cv2.COLOR_YCrCb2BGR)

    # 4. Bloque de visualización (separado del procesamiento principal)
    if verbose:
        plt.figure(figsize=(10, 8))

        # Conversión a RGB exclusivamente para una visualización correcta en
    ↪ Matplotlib
        image_rgb = cv2.cvtColor(image, cv2.COLOR_BGR2RGB)
        out_image_rgb = cv2.cvtColor(out_image, cv2.COLOR_BGR2RGB)

        # Muestra de la imagen original
        plt.subplot(2, 2, 1)
        plt.imshow(image_rgb)
        plt.title('Original image')
        plt.axis('off')
```

```

    # Muestra de la imagen procesada
    plt.subplot(2, 2, 2)
    plt.imshow(out_image_rgb)
    plt.title('Equalized histogram image')
    plt.axis('off')

    # Histograma original (usando el canal Y de luminancia)
    orig_y = cv2.cvtColor(image, cv2.COLOR_BGR2YCrCb)[: , : , 0]
    plt.subplot(2, 2, 3)
    plt.hist(orig_y.ravel(), bins=256, range=[0, 256])
    plt.title('Original histogram (Y channel)')

    # Histograma ecualizado
    plt.subplot(2, 2, 4)
    plt.hist(ycrCb_image[: , : , 0].ravel(), bins=256, range=[0, 256])
    plt.title('Equalized histogram (Y channel)')

    plt.tight_layout()
    plt.show()

    return out_image

```

```

[4]: # Read image
image = cv2.imread(images_path + 'batalla.jpg',-1)

# Equalize its histogram
interactive(equalize_chart, image=fixed(image), verbose=fixed(True))

```

```
interactive(children=(Output(),), _dom_classes=('widget-interact',))
```

Thanks to the incredible power of histogram equalization, we've boosted the contrast, revealing the enemy army in all its glorious gloom. No more squinting into the blizzard, Lord Commander! Those White Walkers thought they were being clever, but our pixel-manipulation prowess has exposed their icy faces. Now we know exactly who's coming for tea (and a battle!), and we can finally tell if that's a direwolf or just a really fluffy snowdrift. Victory is practically in the bag!

0.0.2 SMOOTHING

Even after enhancing contrast with histogram equalization, images can suffer from noise, which obscures important details. To address this, we can apply smoothing techniques like the Gaussian filter. Smoothing helps to reduce random variations in image intensity, making features more discernible without losing too much detail. This initial step of noise reduction can significantly improve the quality of subsequent image processing operations.

```

[5]: def gaussian_filter(image, w_kernel, sigma, verbose=False):
    """ Applies Gaussian filter to an image and display it.

```

```

    Args:
        image: Input image
        w_kernel: Kernel aperture size
        sigma: standard deviation of Gaussian distribution
        verbose: Only show images if this is True

    Returns:
        smoothed_img: smoothed image
"""
# Write your code here!

# Create kernel using separability property
s = sigma
w = w_kernel
kernel_1D = np.float32([1 / (s * np.sqrt(2 * np.pi)) * np.exp(-(z**2) / (2*
↪ s**2)) for z in range(-w,w+1))] # Evaluate the gaussian in "expression"
vertical_kernel = kernel_1D.reshape(2*w+1,1) # Reshape it as a matrix with
↪ just one column
horizontal_kernel = kernel_1D.reshape(1,2*w+1) # Reshape it as a matrix
↪ with just one row
kernel = signal.convolve2d(vertical_kernel, horizontal_kernel) # Get the 2D
↪ kernel

# Convolve image and kernel
smoothed_img = cv2.filter2D(image,-1,kernel)

if verbose:
    plt.figure(figsize=(10, 5)) # Set a good figure size
    # Show the initial image
    plt.subplot(121)
    plt.imshow(image, cmap='gray')
    plt.title('Noisy')
    plt.axis("off")

    # Show the resultant one
    plt.subplot(122)
    plt.imshow(smoothed_img, cmap='gray')
    plt.title('Gaussian filter')
    plt.axis("off")
    plt.tight_layout() # Adjust layout to prevent overlapping titles/labels
    plt.show() # Display the plots

return smoothed_img

```

```

[6]: # Interact with the kernel size and the sigma value
noisy_img = cv2.imread(images_path + 'smoothing_fog.jpeg', 0)

```

```
interactive(gaussian_filter, image=fixed(noisy_img), w_kernel=(0,5,1),  
↳sigma=(1,3,0.3), verbose=fixed(True))
```

```
interactive(children=(IntSlider(value=2, description='w_kernel', max=5),  
↳FloatSlider(value=1.9, description='s...
```

0.0.3 CLAHE

After addressing general image noise with smoothing, we can now refine the contrast enhancement using more advanced techniques. While global histogram equalization can introduce artifacts, Contrast Limited Adaptive Histogram Equalization (CLAHE) offers a more localized approach. CLAHE divides the image into small regions, or ‘tiles,’ and performs equalization on each one, limiting the contrast to avoid over-amplification of noise. This method provides a better quality picture by enhancing contrast adaptively, making details clearer, especially in areas that might still be too dark or too bright.

```
[7]: import cv2  
import matplotlib.pyplot as plt  
import numpy as np  
import ipywidgets as widgets  
from ipywidgets import interactive, fixed  
  
def clahe_chart(image, clip_limit=2.0, tile_grid_size=(8,8), verbose=False):  
    """ Applies CLAHE to an image and shows the result.  
  
    Args:  
        image: Input image  
        clip_limit: Threshold for contrast limiting  
        tile_grid_size: Size of grid for histogram equalization  
        verbose: Only show images if this is True  
  
    Returns:  
        out_image: CLAHE enhanced image  
    """  
  
    # Transform image to YCrCb color space  
    image = cv2.cvtColor(image, cv2.COLOR_BGR2YCrCb)  
    out_image = np.copy(image)  
  
    # Create CLAHE object  
    clahe = cv2.createCLAHE(clipLimit=clip_limit, tileGridSize=tile_grid_size)  
  
    # Apply CLAHE to first band of the YCrCb image  
    out_image[:, :, 0] = clahe.apply(out_image[:, :, 0])  
  
    if verbose:  
        plt.figure(figsize=(15, 10)) # Increased figure size for better  
↳visibility
```

```

# Apply global histogram equalization for comparison
equalized_image = np.copy(image)
equalized_image[:, :, 0] = cv2.equalizeHist(equalized_image[:, :, 0])

# Plot histograms
plt.subplot(2,3,4)
plt.hist(image[:, :, 0].ravel(), 256, range=[0, 256])
plt.title('Original histogram')

plt.subplot(2,3,5)
plt.hist(equalized_image[:, :, 0].ravel(), 256, range=[0, 256])
plt.title('Equalized histogram')

plt.subplot(2,3,6)
plt.hist(out_image[:, :, 0].ravel(), 256, range=[0, 256])
plt.title('CLAHE histogram')

# Reconvert images to RGB
image = cv2.cvtColor(image, cv2.COLOR_YCrCb2RGB)
equalized_image = cv2.cvtColor(equalized_image, cv2.COLOR_YCrCb2RGB)
out_image = cv2.cvtColor(out_image, cv2.COLOR_YCrCb2RGB)

# Show the images
plt.subplot(2,3,1)
plt.imshow(image)
plt.title('Original Image')

plt.subplot(2,3,2)
plt.imshow(equalized_image)
plt.title('Global equalization')

plt.subplot(2,3,3)
plt.imshow(out_image)
plt.title('CLAHE')

plt.tight_layout()
plt.show()

return out_image

```

```

[8]: #Create widget object
clip_widget = widgets.FloatSlider(value=2.0, min=1.0, max=10.0, step=0.5,
    ↪description='Clip limit:')

# Read image
image = cv2.imread(images_path + 'batalla.jpg')

```

```
# Interact with your code!
interactive(clahe_chart, image=fixed(image), clip_limit=clip_widget,
↳tile_grid_size=fixed((8,8)), verbose=fixed(True))
```

```
interactive(children=(FloatSlider(value=2.0, description='Clip limit:', max=10.
↳0, min=1.0, step=0.5), Output()...
```

Although we can see aour enemies, the equalized image presents noise in the background and we can get a better quality picture with another image enhancement techniques, such as CLAHE(Contrast Limited Adaptive Histogram Equalization).Clahe will let us limit contrast enhancement and avoid noise amplification We are going to divide the image into small regions called tiles and we are going to equalize each tile locally using a **clipLimit** before calculating the cumulative distribution function (CDF).

#

CHAPTER II: THE EDGES OF THE LONG NIGHT

The ravens keep sending us images from beyond the Wall. However, due to the heavy fog and thick snow, our scouts cannot easily distinguish where the background ends and an enemy begins. To mount an effective watch from the towers of Winterfell, we need an edge detection system to help us recognize enemies, weapons, or incoming siege structures.

0.0.4 DIRECTIONAL EDGE DETECTION (SOBEL)

For this technique, we are going to work with an image of Winterfell. Edges correspond to sharp transitions in pixel intensity. A difference must be significant enough to be considered a true boundary.

There are several edge detection methods, but we will start with first-derivative operators. By convolving the image with Sobel kernels (G_x and G_y), we can calculate the horizontal and vertical gradients. This allows us to estimate the boundary directions and combine them to compute the total gradient magnitude.

```
[11]: def edge_detection_chart(image, kernel_h, kernel_v, threshold, verbose=False):
    """ Computed the gradient of the image, binarizes and display it.

    Args:
        image: Input image
        kernel_h: kernel for horizontal derivative
        kernel_v: kernel for vertical derivative
        threshold: threshold value for binarization
        verbose: Only show images if this is True

    Returns:
        edges: edges binary image
    """

    # Compute derivatives
    d_h = cv2.filter2D(image, cv2.CV_16S, kernel_h)
```

```

d_v = cv2.filter2D(image, cv2.CV_16S, kernel_v)

# Compute gradient
gradient_image = np.add(np.absolute(d_h), np.absolute(d_v))

# Normalize gradient
norm_gradient = cv2.normalize(gradient_image, None, 0, 255, cv2.NORM_MINMAX)

# Threshold to get edges
ret, edges = cv2.threshold(norm_gradient, threshold, 255, cv2.THRESH_BINARY)

if verbose:
    # Set dark style and adjust figure size
    with plt.style.context('dark_background'):
        plt.figure(figsize=(15, 5))

        # Show the initial image
        plt.subplot(131)
        plt.imshow(image, cmap='gray')
        plt.title('Original image')
        plt.axis('off')

        # Show the gradient image
        plt.subplot(132)
        plt.imshow(gradient_image, cmap='gray')
        plt.title('Gradient image')
        plt.axis('off')

        # Show edges image
        plt.subplot(133)
        plt.imshow(edges, cmap='gray')
        plt.title('Edges detected')
        plt.axis('off')

        plt.tight_layout()
        plt.show()

return edges

```

```

[12]: # Read image
image = cv2.imread(images_path + 'winterfell_blizzard.jpg', 0)

# Define kernel (Sobel)

kernel_h = np.array([[1,0,-1],[2,0,-2],[1,0,-1]])*1/4
kernel_v = np.array([[-1,-2,-1],[0,0,0],[1,2,1]])*1/4

```

```
interactive( edge_detection_chart, image=fixed(image),  
    ↪kernel_h=fixed(kernel_h), kernel_v=fixed(kernel_v), threshold=(0,255,10),  
    ↪verbose=fixed(True))
```

```
interactive(children=(IntSlider(value=120, description='threshold', max=255,  
    ↪step=10), Output()), _dom_classes=...
```

As we can see in the original image, the edges transition from dark pixels to brighter ones. This is the reason why we need to use a lower threshold. In this case, a value of 30 works well because it allows us to detect the outlines of the structures and terrain perfectly.

0.0.5 The Lookout's Dilemma: Single-Threshold Edge Detection

0.0.6 CANNY EDGE DETECTION

To overcome the “Lookout's Dilemma” of using a single global threshold, we are going to use the Canny Edge detection, with two thresholds.

The Canny detector consists of four main steps: 1. **Noise Reduction**: Applies a 5x5 Gaussian filter to smooth the image and remove noise. 2. **Intensity Gradient**: Calculates the horizontal and vertical derivatives to find the gradient magnitude and direction. 3. **Non-Maximum Suppression**: Thins out the edges by suppressing pixels that are not local maxima in the direction of the gradient. 4. **Hysteresis Thresholding**: Uses two threshold values (**threshold1** and **threshold2**): - Pixels with a gradient intensity above **threshold2** are classified as strong edges (sure edges). - Pixels below **threshold1** are discarded. - Pixels between these two thresholds are only kept if they are connected to a strong edge.

```
[13]: def canny_edge_chart(image, threshold1, threshold2, verbose=False):  
    """ Applies Canny edge detection and displays the result with a dark_  
    ↪background style.  
    """  
  
    # Apply Canny algorithm  
    edges = cv2.Canny(image, threshold1, threshold2)  
  
    if verbose:  
        with plt.style.context('dark_background'):  
            plt.figure(figsize=(12, 6))  
  
            plt.subplot(121)  
            plt.imshow(image, cmap='gray')  
            plt.title('Original Image')  
            plt.axis('off')  
  
            plt.subplot(122)  
            plt.imshow(edges, cmap='gray')  
            plt.title('Canny Edges (Hysteresis)')  
            plt.axis('off')  
  
            plt.tight_layout()
```

```
plt.show()

return edges
```

```
[14]: # Read the same winterfell image
image_winterfell = cv2.imread(images_path + 'winterfell_blizzard.jpg', 0)

# Set up interactive sliders for both thresholds
interactive(canny_edge_chart,
            image=fixed(image_winterfell),
            threshold1=widgets.IntSlider(value=50, min=0, max=255, step=5,
↳description='Threshold 1 (Min):'),
            threshold2=widgets.IntSlider(value=150, min=0, max=255, step=5,
↳description='Threshold 2 (Max):'),
            verbose=fixed(True))
```

```
interactive(children=(IntSlider(value=50, description='Threshold 1 (Min):',
↳max=255, step=5), IntSlider(value=...
```

0.0.7 COMPARATIVE CONCLUSION

Mientras que **Sobel** es un método rápido y sencillo que detecta transiciones básicas pero es muy sensible al ruido y genera bordes gruesos y fragmentados, el algoritmo de **Canny** ofrece un resultado muy superior. Al integrar un filtro Gaussiano previo para reducir el ruido, realizar supresión de no-máximos para adelgazar los bordes y aplicar un doble umbral (histéresis), Canny logra trazar contornos limpios, continuos y precisos sin verse afectado por el caos de la ventisca.

0.0.8 EXPORTAR A PDF SIN RECORTES (RECOMENDADO)

Ejecuta la celda inferior para instalar las herramientas de LaTeX y convertir directamente este cuaderno a un PDF perfectamente maquetado.

```
[15]: # Instalar dependencias del sistema para la conversión a PDF
!apt-get update
!apt-get install -y texlive-xetex texlive-fonts-recommended
↳texlive-plain-generic
```

```
Hit:1 http://archive.ubuntu.com/ubuntu noble InRelease
Get:2 https://cloud.r-project.org/bin/linux/ubuntu noble-cran40/ InRelease
[3,631 B]
Get:3 https://cli.github.com/packages stable InRelease [3,917 B]
Get:4 http://archive.ubuntu.com/ubuntu noble-updates InRelease [126 kB]
Get:5 http://security.ubuntu.com/ubuntu noble-security InRelease [126 kB]
Get:6 https://r2u.stat.illinois.edu/ubuntu noble InRelease [9,161 B]
Get:7 http://archive.ubuntu.com/ubuntu noble-backports InRelease [126 kB]
Get:8 https://cloud.r-project.org/bin/linux/ubuntu noble-cran40/ Packages [73.2
kB]
```

```

Get:9 https://cli.github.com/packages stable/main amd64 Packages [356 B]
Get:10 https://ppa.launchpadcontent.net/deadsnakes/ppa/ubuntu noble InRelease
[17.8 kB]
Get:11 https://r2u.stat.illinois.edu/ubuntu noble/main all Packages [10.3 MB]
Get:12 http://security.ubuntu.com/ubuntu noble-security/universe amd64 Packages
[1,542 kB]
Get:13 http://archive.ubuntu.com/ubuntu noble-updates/main amd64 Packages [1,667
kB]
Get:14 http://security.ubuntu.com/ubuntu noble-security/main amd64 Packages
[1,315 kB]
Get:15 http://security.ubuntu.com/ubuntu noble-security/restricted amd64
Packages [1,943 kB]
Get:16 https://r2u.stat.illinois.edu/ubuntu noble/main amd64 Packages [3,036 kB]
Get:17 http://archive.ubuntu.com/ubuntu noble-updates/restricted amd64 Packages
[2,112 kB]
Get:18 http://archive.ubuntu.com/ubuntu noble-updates/universe amd64 Packages
[2,162 kB]
Get:19 http://archive.ubuntu.com/ubuntu noble-backports/universe amd64 Packages
[36.0 kB]
Get:20 http://archive.ubuntu.com/ubuntu noble-backports/main amd64 Packages
[49.0 kB]
Get:21 https://ppa.launchpadcontent.net/deadsnakes/ppa/ubuntu noble/main amd64
Packages [52.1 kB]
Fetched 24.7 MB in 3s (9,227 kB/s)
Reading package lists... Done
W: Skipping acquire of configured file 'main/source/Sources' as repository
'https://r2u.stat.illinois.edu/ubuntu noble InRelease' does not seem to provide
it (sources.list entry misspelt?)
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
The following additional packages will be installed:
  at-spi2-common at-spi2-core default-jre default-jre-headless dvisvgm
  fonts-dejavu-core fonts-dejavu-extra fonts-dejavu-mono fonts-droid-fallback
  fonts-lato fonts-lmodern fonts- noto-mono fonts-texgyre fonts-texgyre-math
  fonts-urw-base35 gsettings-desktop-schemas libapache-pom-java
  libatk-bridge2.0-0t64 libatk-wrapper-java libatk-wrapper-java-jni
  libatk1.0-0t64 libatspi2.0-0t64 libbit-vector-perl libcarp-clan-perl
  libcommons-logging-java libcommons-parent-java libcrypt-rc4-perl
  libdate-calc-perl libdate-calc-xs-perl libdate-manip-perl
  libdigest-perl-md5-perl libfontbox-java libgs-common libgs10 libgs10-common
  libgtk-3-0t64 libgtk-3-bin libgtk-3-common libgumbo2 libidn12 libijs-0.35
  libio-stringy-perl libjbig2dec0 libjcode-pm-perl libkpathsea6 libmujs3
  libole-storage-lite-perl libparse-recdescent-perl libpdfbox-java libpotrace0
  libptexenc1 librsvg2-common libruby libruby3.2
  libspreadsheet-parseexcel-perl libspreadsheet-writeexcel-perl libsynchronex2
  libteckit0 libtexlua53-5 libunicode-map-perl libwoff1 libxcomposite1
  libxtst6 libxxf86dga1 libzzip-0-13t64 lmodern mupdf-tools

```

openjdk-21-jdk-headless openjdk-21-jre openjdk-21-jre-headless poppler-data
preview-latex-style rake ruby ruby-net-telnet ruby-rubygems ruby-sdbm
ruby-webrick ruby-xmlrpc ruby3.2 rubygems-integration session-migration
tlutils teckit tex-common tex-gyre texlive-base texlive-binaries
texlive-latex-base texlive-latex-extra texlive-latex-recommended
texlive-pictures tipa x11-utils xfonts-encodings xfonts-utils

Suggested packages:

fonts-noto fonts-freefont-otf | fonts-freefont-ttf libavalon-framework-java
libexcalibur-logkit-java liblog4j1.2-java gvfs libwww-perl openjdk-21-demo
openjdk-21-source libnss-mdns fonts-ipafont-gothic fonts-ipafont-mincho
fonts-wqy-microhei | fonts-wqy-zenhei fonts-indic poppler-utils ghostscript
fonts-japanese-mincho | fonts-ipafont-mincho fonts-japanese-gothic
| fonts-ipafont-gothic fonts-arphic-ukai fonts-arphic-uming fonts-nanum ri
ruby-dev bundler debhelper gv | postscript-viewer perl-tk xpdf | pdf-viewer
xzdec texlive-binaries-sse2 hintview texlive-fonts-recommended-doc
texlive-latex-base-doc wp2latex python3-pygments icc-profiles
libfile-which-perl texlive-latex-extra-doc texlive-science
texlive-latex-recommended-doc texlive-luatex texlive-pstricks dot2tex prerex
texlive-pictures-doc vprerex tipa-doc mesa-utils

Recommended packages:

luit

The following NEW packages will be installed:

at-spi2-common at-spi2-core default-jre default-jre-headless dvisvgm
fonts-dejavu-core fonts-dejavu-extra fonts-dejavu-mono fonts-droid-fallback
fonts-lato fonts-lmodern fonts-noto-mono fonts-texgyre fonts-texgyre-math
fonts-urw-base35 gsettings-desktop-schemas libapache-pom-java
libatk-bridge2.0-0t64 libatk-wrapper-java libatk-wrapper-java-jni
libatk1.0-0t64 libatspi2.0-0t64 libbit-vector-perl libcarp-clan-perl
libcommons-logging-java libcommons-parent-java libcrypt-rc4-perl
libdate-calc-perl libdate-calc-xs-perl libdate-manip-perl
libdigest-perl-md5-perl libfontbox-java libgs-common libgs10 libgs10-common
libgtk-3-0t64 libgtk-3-bin libgtk-3-common libgumbo2 libidn12 libijs-0.35
libio-stringy-perl libjbig2dec0 libjcode-pm-perl libkpathsea6 libmujs3
libole-storage-lite-perl libparse-recdescent-perl libpdfbox-java libpotrace0
libptexenc1 librsvg2-common libruby libruby3.2
libspreadsheet-parseexcel-perl libspreadsheet-writeexcel-perl libsynctex2
libteckit0 libtexlua53-5 libunicode-map-perl libwoff1 libxcomposite1
libxtst6 libxxf86dga1 libzip-0-13t64 lmodern mupdf-tools openjdk-21-jre
poppler-data preview-latex-style rake ruby ruby-net-telnet ruby-rubygems
ruby-sdbm ruby-webrick ruby-xmlrpc ruby3.2 rubygems-integration
session-migration tlutils teckit tex-common tex-gyre texlive-base
texlive-binaries texlive-fonts-recommended texlive-latex-base
texlive-latex-extra texlive-latex-recommended texlive-pictures
texlive-plain-generic texlive-xetex tipa x11-utils xfonts-encodings
xfonts-utils

The following packages will be upgraded:

openjdk-21-jdk-headless openjdk-21-jre-headless

2 upgraded, 97 newly installed, 0 to remove and 55 not upgraded.

Need to get 383 MB of archives.

After this operation, 739 MB of additional disk space will be used.

Get:1 <http://archive.ubuntu.com/ubuntu/noble/main/amd64/fonts-droid-fallback> all 1:6.0.1r16-1.1build1 [1,805 kB]

Get:2 <http://archive.ubuntu.com/ubuntu/noble/main/amd64/fonts-lato> all 2.015-1 [2,781 kB]

Get:3 <http://archive.ubuntu.com/ubuntu/noble/main/amd64/poppler-data> all 0.4.12-1 [2,060 kB]

Get:4 <http://archive.ubuntu.com/ubuntu/noble/main/amd64/at-spi2-common> all 2.52.0-1build1 [8,674 B]

Get:5 <http://archive.ubuntu.com/ubuntu/noble/main/amd64/libatspi2.0-0t64> amd64 2.52.0-1build1 [80.5 kB]

Get:6 <http://archive.ubuntu.com/ubuntu/noble/main/amd64/libxtst6> amd64 2:1.2.3-1.1build1 [12.6 kB]

Get:7 <http://archive.ubuntu.com/ubuntu/noble/main/amd64/session-migration> amd64 0.3.9build1 [9,034 B]

Get:8 <http://archive.ubuntu.com/ubuntu/noble-updates/main/amd64/gsettings-desktop-schemas> all 46.1-0ubuntu1 [35.6 kB]

Get:9 <http://archive.ubuntu.com/ubuntu/noble/main/amd64/at-spi2-core> amd64 2.52.0-1build1 [56.6 kB]

Get:10 <http://archive.ubuntu.com/ubuntu/noble-updates/main/amd64/openjdk-21-jdk-headless> amd64 21.0.12.1+1-1~24.04.4 [82.8 MB]

Get:11 <http://archive.ubuntu.com/ubuntu/noble-updates/main/amd64/openjdk-21-jre-headless> amd64 21.0.12.1+1-1~24.04.4 [46.5 MB]

Get:12 <http://archive.ubuntu.com/ubuntu/noble/main/amd64/default-jre-headless> amd64 2:1.21-75+exp1 [3,094 B]

Get:13 <http://archive.ubuntu.com/ubuntu/noble/main/amd64/libatk1.0-0t64> amd64 2.52.0-1build1 [55.3 kB]

Get:14 <http://archive.ubuntu.com/ubuntu/noble/main/amd64/libatk-bridge2.0-0t64> amd64 2.52.0-1build1 [66.0 kB]

Get:15 <http://archive.ubuntu.com/ubuntu/noble/main/amd64/libxcomposite1> amd64 1:0.4.5-1build3 [6,320 B]

Get:16 <http://archive.ubuntu.com/ubuntu/noble-updates/main/amd64/libgtk-3-common> all 3.24.41-4ubuntu1.3 [1,426 kB]

Get:17 <http://archive.ubuntu.com/ubuntu/noble-updates/main/amd64/libgtk-3-0t64> amd64 3.24.41-4ubuntu1.3 [2,913 kB]

Get:18 <http://archive.ubuntu.com/ubuntu/noble-updates/main/amd64/openjdk-21-jre> amd64 21.0.12.1+1-1~24.04.4 [233 kB]

Get:19 <http://archive.ubuntu.com/ubuntu/noble/main/amd64/default-jre> amd64 2:1.21-75+exp1 [922 B]

Get:20 <http://archive.ubuntu.com/ubuntu/noble/main/amd64/xfonts-encodings> all 1:1.0.5-0ubuntu2 [578 kB]

Get:21 <http://archive.ubuntu.com/ubuntu/noble/main/amd64/xfonts-utils> amd64 1:7.7+6build3 [94.4 kB]

Get:22 <http://archive.ubuntu.com/ubuntu/noble/main/amd64/fonts-urw-base35> all 20200910-8 [11.0 MB]

Get:23 <http://archive.ubuntu.com/ubuntu/noble-updates/main/amd64/libgs-common> all 10.02.1~dfsg1-0ubuntu7.9 [176 kB]

Get:24 <http://archive.ubuntu.com/ubuntu> noble-updates/main amd64 libgs10-common all 10.02.1~dfsg1-0ubuntu7.9 [488 kB]
Get:25 <http://archive.ubuntu.com/ubuntu> noble-updates/main amd64 libidn12 amd64 1.42-1ubuntu0.1 [56.1 kB]
Get:26 <http://archive.ubuntu.com/ubuntu> noble/main amd64 libijs-0.35 amd64 0.35-15.1build1 [15.3 kB]
Get:27 <http://archive.ubuntu.com/ubuntu> noble-updates/main amd64 libjbig2dec0 amd64 0.20-1ubuntu0.24.04.1 [65.2 kB]
Get:28 <http://archive.ubuntu.com/ubuntu> noble-updates/main amd64 libgs10 amd64 10.02.1~dfsg1-0ubuntu7.9 [3,898 kB]
Get:29 <http://archive.ubuntu.com/ubuntu> noble/main amd64 libkpathsea6 amd64 2023.20230311.66589-9build3 [63.0 kB]
Get:30 <http://archive.ubuntu.com/ubuntu> noble/universe amd64 libpotrace0 amd64 1.16-2build1 [17.7 kB]
Get:31 <http://archive.ubuntu.com/ubuntu> noble/main amd64 libwoff1 amd64 1.0.2-2build1 [45.3 kB]
Get:32 <http://archive.ubuntu.com/ubuntu> noble/universe amd64 dvisvgm amd64 3.2.1+ds-1build1 [1,042 kB]
Get:33 <http://archive.ubuntu.com/ubuntu> noble/main amd64 fonts-dejavu-mono all 2.37-8 [502 kB]
Get:34 <http://archive.ubuntu.com/ubuntu> noble/main amd64 fonts-dejavu-core all 2.37-8 [835 kB]
Get:35 <http://archive.ubuntu.com/ubuntu> noble/main amd64 fonts-dejavu-extra all 2.37-8 [1,947 kB]
Get:36 <http://archive.ubuntu.com/ubuntu> noble/universe amd64 fonts-lmodern all 2.005-1 [4,799 kB]
Get:37 <http://archive.ubuntu.com/ubuntu> noble/main amd64 fonts-noto-mono all 20201225-2 [435 kB]
Get:38 <http://archive.ubuntu.com/ubuntu> noble/universe amd64 fonts-texgyre all 20180621-6 [8,350 kB]
Get:39 <http://archive.ubuntu.com/ubuntu> noble/universe amd64 fonts-texgyre-math all 20180621-6 [2,373 kB]
Get:40 <http://archive.ubuntu.com/ubuntu> noble/universe amd64 libapache-pom-java all 29-2 [5,284 B]
Get:41 <http://archive.ubuntu.com/ubuntu> noble/main amd64 libxxf86dga1 amd64 2:1.1.5-1build1 [11.6 kB]
Get:42 <http://archive.ubuntu.com/ubuntu> noble/main amd64 x11-utils amd64 7.7+6build2 [189 kB]
Get:43 <http://archive.ubuntu.com/ubuntu> noble/main amd64 libatk-wrapper-java all 0.40.0-3build2 [54.3 kB]
Get:44 <http://archive.ubuntu.com/ubuntu> noble/main amd64 libatk-wrapper-java-jni amd64 0.40.0-3build2 [46.4 kB]
Get:45 <http://archive.ubuntu.com/ubuntu> noble/universe amd64 libcarp-clan-perl all 6.08-2 [11.9 kB]
Get:46 <http://archive.ubuntu.com/ubuntu> noble/universe amd64 libbit-vector-perl amd64 7.4-3build1 [139 kB]
Get:47 <http://archive.ubuntu.com/ubuntu> noble/universe amd64 libcommons-parent-java all 56-1 [10.7 kB]

Get:48 <http://archive.ubuntu.com/ubuntu> noble/universe amd64 libcommons-logging-java all 1.3.0-1ubuntu1 [63.8 kB]
Get:49 <http://archive.ubuntu.com/ubuntu> noble/universe amd64 libcrypt-rc4-perl all 2.02-5 [7,064 B]
Get:50 <http://archive.ubuntu.com/ubuntu> noble/universe amd64 libdate-calc-perl all 6.4-2 [187 kB]
Get:51 <http://archive.ubuntu.com/ubuntu> noble/universe amd64 libdate-calc-xs-perl amd64 6.4-2build4 [46.8 kB]
Get:52 <http://archive.ubuntu.com/ubuntu> noble/main amd64 libdate-manip-perl all 6.95-1 [923 kB]
Get:53 <http://archive.ubuntu.com/ubuntu> noble/universe amd64 libdigest-perl-md5-perl all 1.9-5 [11.1 kB]
Get:54 <http://archive.ubuntu.com/ubuntu> noble-updates/main amd64 libgtk-3-bin amd64 3.24.41-4ubuntu1.3 [73.9 kB]
Get:55 <http://archive.ubuntu.com/ubuntu> noble/universe amd64 libgumbo2 amd64 0.12.0+dfsg-2build1 [126 kB]
Get:56 <http://archive.ubuntu.com/ubuntu> noble/main amd64 libio-stringy-perl all 2.111-3 [55.8 kB]
Get:57 <http://archive.ubuntu.com/ubuntu> noble/universe amd64 libjcode-pm-perl amd64 2.06-1.1build1 [26.8 kB]
Get:58 <http://archive.ubuntu.com/ubuntu> noble/universe amd64 libmujs3 amd64 1.3.3-3build2 [134 kB]
Get:59 <http://archive.ubuntu.com/ubuntu> noble/universe amd64 libole-storage-lite-perl all 0.22-1 [21.6 kB]
Get:60 <http://archive.ubuntu.com/ubuntu> noble/universe amd64 libparse-recdescent-perl all 1.967015+dfsg-4 [139 kB]
Get:61 <http://archive.ubuntu.com/ubuntu> noble/main amd64 libptexenc1 amd64 2023.20230311.66589-9build3 [40.4 kB]
Get:62 <http://archive.ubuntu.com/ubuntu> noble/main amd64 librsvg2-common amd64 2.58.0+dfsg-1build1 [11.8 kB]
Get:63 <http://archive.ubuntu.com/ubuntu> noble/main amd64 rubygems-integration all 1.18 [5,336 B]
Get:64 <http://archive.ubuntu.com/ubuntu> noble-updates/main amd64 ruby3.2 amd64 3.2.3-1ubuntu0.24.04.8 [50.7 kB]
Get:65 <http://archive.ubuntu.com/ubuntu> noble/main amd64 ruby-rubygems all 3.4.20-1 [238 kB]
Get:66 <http://archive.ubuntu.com/ubuntu> noble/main amd64 ruby amd64 1:3.2~ubuntu1 [3,466 B]
Get:67 <http://archive.ubuntu.com/ubuntu> noble/main amd64 rake all 13.0.6-3 [61.6 kB]
Get:68 <http://archive.ubuntu.com/ubuntu> noble/main amd64 ruby-net-telnet all 0.2.0-1 [13.3 kB]
Get:69 <http://archive.ubuntu.com/ubuntu> noble-updates/main amd64 ruby-webrick all 1.8.1-1ubuntu0.2 [53.0 kB]
Get:70 <http://archive.ubuntu.com/ubuntu> noble/main amd64 ruby-xmlrpc all 0.3.2-2 [24.8 kB]
Get:71 <http://archive.ubuntu.com/ubuntu> noble/main amd64 ruby-sdbm amd64 1.0.0-5build4 [16.2 kB]

Get:72 <http://archive.ubuntu.com/ubuntu> noble-updates/main amd64 libruby3.2 amd64 3.2.3-1ubuntu0.24.04.8 [5,342 kB]
 Get:73 <http://archive.ubuntu.com/ubuntu> noble/main amd64 libruby amd64 1:3.2~ubuntu1 [4,694 B]
 Get:74 <http://archive.ubuntu.com/ubuntu> noble/universe amd64 libspreadsheet-writeexcel-perl all 2.40-4 [497 kB]
 Get:75 <http://archive.ubuntu.com/ubuntu> noble/universe amd64 libunicode-map-perl amd64 0.112-13build4 [203 kB]
 Get:76 <http://archive.ubuntu.com/ubuntu> noble/universe amd64 libspreadsheet-parseexcel-perl all 0.6600-1 [144 kB]
 Get:77 <http://archive.ubuntu.com/ubuntu> noble/main amd64 libsinctex2 amd64 2023.20230311.66589-9build3 [59.6 kB]
 Get:78 <http://archive.ubuntu.com/ubuntu> noble/universe amd64 libteckit0 amd64 2.5.12+ds1-1 [411 kB]
 Get:79 <http://archive.ubuntu.com/ubuntu> noble/main amd64 libtexlua53-5 amd64 2023.20230311.66589-9build3 [123 kB]
 Get:80 <http://archive.ubuntu.com/ubuntu> noble/universe amd64 libzip-0-13t64 amd64 0.13.72+dfsg.1-1.2build1 [28.1 kB]
 Get:81 <http://archive.ubuntu.com/ubuntu> noble/universe amd64 tex-common all 6.18 [32.8 kB]
 Get:82 <http://archive.ubuntu.com/ubuntu> noble/universe amd64 lmodern all 2.005-1 [9,542 kB]
 Get:83 <http://archive.ubuntu.com/ubuntu> noble/universe amd64 mupdf-tools amd64 1.23.10+ds1-1build3 [49.3 MB]
 Get:84 <http://archive.ubuntu.com/ubuntu> noble/universe amd64 preview-latex-style all 13.2-1 [347 kB]
 Get:85 <http://archive.ubuntu.com/ubuntu> noble/main amd64 t1utils amd64 1.41-4build3 [61.3 kB]
 Get:86 <http://archive.ubuntu.com/ubuntu> noble/universe amd64 teckit amd64 2.5.12+ds1-1 [713 kB]
 Get:87 <http://archive.ubuntu.com/ubuntu> noble/universe amd64 tex-gyre all 20180621-6 [6,396 kB]
 Get:88 <http://archive.ubuntu.com/ubuntu> noble/universe amd64 texlive-binaries amd64 2023.20230311.66589-9build3 [8,529 kB]
 Get:89 <http://archive.ubuntu.com/ubuntu> noble/universe amd64 texlive-base all 2023.20240207-1 [21.7 MB]
 Get:90 <http://archive.ubuntu.com/ubuntu> noble/universe amd64 texlive-fonts-recommended all 2023.20240207-1 [4,973 kB]
 Get:91 <http://archive.ubuntu.com/ubuntu> noble/universe amd64 texlive-latex-base all 2023.20240207-1 [1,238 kB]
 Get:92 <http://archive.ubuntu.com/ubuntu> noble/universe amd64 libfontbox-java all 1:1.8.16-5 [208 kB]
 Get:93 <http://archive.ubuntu.com/ubuntu> noble/universe amd64 libpdfbox-java all 1:1.8.16-5 [5,521 kB]
 Get:94 <http://archive.ubuntu.com/ubuntu> noble/universe amd64 texlive-latex-recommended all 2023.20240207-1 [8,826 kB]
 Get:95 <http://archive.ubuntu.com/ubuntu> noble/universe amd64 texlive-pictures all 2023.20240207-1 [16.7 MB]

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Get:96 http://archive.ubuntu.com/ubuntu noble/universe amd64 texlive-latex-extra
all 2023.20240207-1 [19.2 MB]
Get:97 http://archive.ubuntu.com/ubuntu noble/universe amd64 texlive-plain-
generic all 2023.20240207-1 [29.0 MB]
Get:98 http://archive.ubuntu.com/ubuntu noble/universe amd64 tipa all 2:1.3-21
[2,967 kB]
Get:99 http://archive.ubuntu.com/ubuntu noble/universe amd64 texlive-xetex all
2023.20240207-1 [10.8 MB]
Fetched 383 MB in 7s (57.1 MB/s)
Extracting templates from packages: 100%
Preconfiguring packages ...
Selecting previously unselected package fonts-droid-fallback.
(Reading database ... 122809 files and directories currently installed.)
Preparing to unpack .../00-fonts-droid-fallback_1%3a6.0.1r16-1.1build1_all.deb
...
Unpacking fonts-droid-fallback (1:6.0.1r16-1.1build1) ...
Selecting previously unselected package fonts-lato.
Preparing to unpack .../01-fonts-lato_2.015-1_all.deb ...
Unpacking fonts-lato (2.015-1) ...
Selecting previously unselected package poppler-data.
Preparing to unpack .../02-poppler-data_0.4.12-1_all.deb ...
Unpacking poppler-data (0.4.12-1) ...
Selecting previously unselected package at-spi2-common.
Preparing to unpack .../03-at-spi2-common_2.52.0-1build1_all.deb ...
Unpacking at-spi2-common (2.52.0-1build1) ...
Selecting previously unselected package libatspi2.0-0t64:amd64.
Preparing to unpack .../04-libatspi2.0-0t64_2.52.0-1build1_amd64.deb ...
Unpacking libatspi2.0-0t64:amd64 (2.52.0-1build1) ...
Selecting previously unselected package libxtst6:amd64.
Preparing to unpack .../05-libxtst6_2%3a1.2.3-1.1build1_amd64.deb ...
Unpacking libxtst6:amd64 (2:1.2.3-1.1build1) ...
Selecting previously unselected package session-migration.
Preparing to unpack .../06-session-migration_0.3.9build1_amd64.deb ...
Unpacking session-migration (0.3.9build1) ...
Selecting previously unselected package gsettings-desktop-schemas.
Preparing to unpack .../07-gsettings-desktop-schemas_46.1-0ubuntu1_all.deb ...
Unpacking gsettings-desktop-schemas (46.1-0ubuntu1) ...
Selecting previously unselected package at-spi2-core.
Preparing to unpack .../08-at-spi2-core_2.52.0-1build1_amd64.deb ...
Unpacking at-spi2-core (2.52.0-1build1) ...
Preparing to unpack .../09-openjdk-21-jdk-
headless_21.0.12.1+1-1~24.04.4_amd64.deb ...
Unpacking openjdk-21-jdk-headless:amd64 (21.0.12.1+1-1~24.04.4) over
(21.0.12+8-1~24.04) ...
Preparing to unpack .../10-openjdk-21-jre-
headless_21.0.12.1+1-1~24.04.4_amd64.deb ...
Unpacking openjdk-21-jre-headless:amd64 (21.0.12.1+1-1~24.04.4) over
(21.0.12+8-1~24.04) ...

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Selecting previously unselected package default-jre-headless.
 Preparing to unpack .../11-default-jre-headless_2%3a1.21-75+exp1_amd64.deb ...
 Unpacking default-jre-headless (2:1.21-75+exp1) ...
 Selecting previously unselected package libatk1.0-0t64:amd64.
 Preparing to unpack .../12-libatk1.0-0t64_2.52.0-1build1_amd64.deb ...
 Unpacking libatk1.0-0t64:amd64 (2.52.0-1build1) ...
 Selecting previously unselected package libatk-bridge2.0-0t64:amd64.
 Preparing to unpack .../13-libatk-bridge2.0-0t64_2.52.0-1build1_amd64.deb ...
 Unpacking libatk-bridge2.0-0t64:amd64 (2.52.0-1build1) ...
 Selecting previously unselected package libxcomposite1:amd64.
 Preparing to unpack .../14-libxcomposite1_1%3a0.4.5-1build3_amd64.deb ...
 Unpacking libxcomposite1:amd64 (1:0.4.5-1build3) ...
 Selecting previously unselected package libgtk-3-common.
 Preparing to unpack .../15-libgtk-3-common_3.24.41-4ubuntu1.3_all.deb ...
 Unpacking libgtk-3-common (3.24.41-4ubuntu1.3) ...
 Selecting previously unselected package libgtk-3-0t64:amd64.
 Preparing to unpack .../16-libgtk-3-0t64_3.24.41-4ubuntu1.3_amd64.deb ...
 Unpacking libgtk-3-0t64:amd64 (3.24.41-4ubuntu1.3) ...
 Selecting previously unselected package openjdk-21-jre:amd64.
 Preparing to unpack .../17-openjdk-21-jre_21.0.12.1+1-1~24.04.4_amd64.deb ...
 Unpacking openjdk-21-jre:amd64 (21.0.12.1+1-1~24.04.4) ...
 Selecting previously unselected package default-jre.
 Preparing to unpack .../18-default-jre_2%3a1.21-75+exp1_amd64.deb ...
 Unpacking default-jre (2:1.21-75+exp1) ...
 Selecting previously unselected package xfonts-encodings.
 Preparing to unpack .../19-xfonts-encodings_1%3a1.0.5-0ubuntu2_all.deb ...
 Unpacking xfonts-encodings (1:1.0.5-0ubuntu2) ...
 Selecting previously unselected package xfonts-utils.
 Preparing to unpack .../20-xfonts-utils_1%3a7.7+6build3_amd64.deb ...
 Unpacking xfonts-utils (1:7.7+6build3) ...
 Selecting previously unselected package fonts-urw-base35.
 Preparing to unpack .../21-fonts-urw-base35_20200910-8_all.deb ...
 Unpacking fonts-urw-base35 (20200910-8) ...
 Selecting previously unselected package libgs-common.
 Preparing to unpack .../22-libgs-common_10.02.1~dfsg1-0ubuntu7.9_all.deb ...
 Unpacking libgs-common (10.02.1~dfsg1-0ubuntu7.9) ...
 Selecting previously unselected package libgs10-common.
 Preparing to unpack .../23-libgs10-common_10.02.1~dfsg1-0ubuntu7.9_all.deb ...
 Unpacking libgs10-common (10.02.1~dfsg1-0ubuntu7.9) ...
 Selecting previously unselected package libidn12:amd64.
 Preparing to unpack .../24-libidn12_1.42-1ubuntu0.1_amd64.deb ...
 Unpacking libidn12:amd64 (1.42-1ubuntu0.1) ...
 Selecting previously unselected package libijs-0.35:amd64.
 Preparing to unpack .../25-libijs-0.35_0.35-15.1build1_amd64.deb ...
 Unpacking libijs-0.35:amd64 (0.35-15.1build1) ...
 Selecting previously unselected package libjbig2dec0:amd64.
 Preparing to unpack .../26-libjbig2dec0_0.20-1ubuntu0.24.04.1_amd64.deb ...
 Unpacking libjbig2dec0:amd64 (0.20-1ubuntu0.24.04.1) ...

```

Selecting previously unselected package libgs10:amd64.
Preparing to unpack .../27-libgs10_10.02.1~dfsg1-0ubuntu7.9_amd64.deb ...
Unpacking libgs10:amd64 (10.02.1~dfsg1-0ubuntu7.9) ...
Selecting previously unselected package libkpathsea6:amd64.
Preparing to unpack .../28-libkpathsea6_2023.20230311.66589-9build3_amd64.deb
...
Unpacking libkpathsea6:amd64 (2023.20230311.66589-9build3) ...
Selecting previously unselected package libpotrace0:amd64.
Preparing to unpack .../29-libpotrace0_1.16-2build1_amd64.deb ...
Unpacking libpotrace0:amd64 (1.16-2build1) ...
Selecting previously unselected package libwoff1:amd64.
Preparing to unpack .../30-libwoff1_1.0.2-2build1_amd64.deb ...
Unpacking libwoff1:amd64 (1.0.2-2build1) ...
Selecting previously unselected package dvisvgm.
Preparing to unpack .../31-dvisvgm_3.2.1+ds-1build1_amd64.deb ...
Unpacking dvisvgm (3.2.1+ds-1build1) ...
Selecting previously unselected package fonts-dejavu-mono.
Preparing to unpack .../32-fonts-dejavu-mono_2.37-8_all.deb ...
Unpacking fonts-dejavu-mono (2.37-8) ...
Selecting previously unselected package fonts-dejavu-core.
Preparing to unpack .../33-fonts-dejavu-core_2.37-8_all.deb ...
Unpacking fonts-dejavu-core (2.37-8) ...
Selecting previously unselected package fonts-dejavu-extra.
Preparing to unpack .../34-fonts-dejavu-extra_2.37-8_all.deb ...
Unpacking fonts-dejavu-extra (2.37-8) ...
Selecting previously unselected package fonts-lmodern.
Preparing to unpack .../35-fonts-lmodern_2.005-1_all.deb ...
Unpacking fonts-lmodern (2.005-1) ...
Selecting previously unselected package fonts-noto-mono.
Preparing to unpack .../36-fonts-noto-mono_20201225-2_all.deb ...
Unpacking fonts-noto-mono (20201225-2) ...
Selecting previously unselected package fonts-texgyre.
Preparing to unpack .../37-fonts-texgyre_20180621-6_all.deb ...
Unpacking fonts-texgyre (20180621-6) ...
Selecting previously unselected package fonts-texgyre-math.
Preparing to unpack .../38-fonts-texgyre-math_20180621-6_all.deb ...
Unpacking fonts-texgyre-math (20180621-6) ...
Selecting previously unselected package libapache-pom-java.
Preparing to unpack .../39-libapache-pom-java_29-2_all.deb ...
Unpacking libapache-pom-java (29-2) ...
Selecting previously unselected package libxxf86dga1:amd64.
Preparing to unpack .../40-libxxf86dga1_2%3a1.1.5-1build1_amd64.deb ...
Unpacking libxxf86dga1:amd64 (2:1.1.5-1build1) ...
Selecting previously unselected package x11-utils.
Preparing to unpack .../41-x11-utils_7.7+6build2_amd64.deb ...
Unpacking x11-utils (7.7+6build2) ...
Selecting previously unselected package libatk-wrapper-java.
Preparing to unpack .../42-libatk-wrapper-java_0.40.0-3build2_all.deb ...

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Unpacking libatk-wrapper-java (0.40.0-3build2) ...
Selecting previously unselected package libatk-wrapper-java-jni:amd64.
Preparing to unpack .../43-libatk-wrapper-java-jni_0.40.0-3build2_amd64.deb ...
Unpacking libatk-wrapper-java-jni:amd64 (0.40.0-3build2) ...
Selecting previously unselected package libcarp-clan-perl.
Preparing to unpack .../44-libcarp-clan-perl_6.08-2_all.deb ...
Unpacking libcarp-clan-perl (6.08-2) ...
Selecting previously unselected package libbit-vector-perl:amd64.
Preparing to unpack .../45-libbit-vector-perl_7.4-3build1_amd64.deb ...
Unpacking libbit-vector-perl:amd64 (7.4-3build1) ...
Selecting previously unselected package libcommons-parent-java.
Preparing to unpack .../46-libcommons-parent-java_56-1_all.deb ...
Unpacking libcommons-parent-java (56-1) ...
Selecting previously unselected package libcommons-logging-java.
Preparing to unpack .../47-libcommons-logging-java_1.3.0-1ubuntu1_all.deb ...
Unpacking libcommons-logging-java (1.3.0-1ubuntu1) ...
Selecting previously unselected package libcrypt-rc4-perl.
Preparing to unpack .../48-libcrypt-rc4-perl_2.02-5_all.deb ...
Unpacking libcrypt-rc4-perl (2.02-5) ...
Selecting previously unselected package libdate-calc-perl.
Preparing to unpack .../49-libdate-calc-perl_6.4-2_all.deb ...
Unpacking libdate-calc-perl (6.4-2) ...
Selecting previously unselected package libdate-calc-xs-perl:amd64.
Preparing to unpack .../50-libdate-calc-xs-perl_6.4-2build4_amd64.deb ...
Unpacking libdate-calc-xs-perl:amd64 (6.4-2build4) ...
Selecting previously unselected package libdate-manip-perl.
Preparing to unpack .../51-libdate-manip-perl_6.95-1_all.deb ...
Unpacking libdate-manip-perl (6.95-1) ...
Selecting previously unselected package libdigest-perl-md5-perl.
Preparing to unpack .../52-libdigest-perl-md5-perl_1.9-5_all.deb ...
Unpacking libdigest-perl-md5-perl (1.9-5) ...
Selecting previously unselected package libgtk-3-bin.
Preparing to unpack .../53-libgtk-3-bin_3.24.41-4ubuntu1.3_amd64.deb ...
Unpacking libgtk-3-bin (3.24.41-4ubuntu1.3) ...
Selecting previously unselected package libgumbo2:amd64.
Preparing to unpack .../54-libgumbo2_0.12.0+dfsg-2build1_amd64.deb ...
Unpacking libgumbo2:amd64 (0.12.0+dfsg-2build1) ...
Selecting previously unselected package libio-stringy-perl.
Preparing to unpack .../55-libio-stringy-perl_2.111-3_all.deb ...
Unpacking libio-stringy-perl (2.111-3) ...
Selecting previously unselected package libjcode-pm-perl.
Preparing to unpack .../56-libjcode-pm-perl_2.06-1.1build1_amd64.deb ...
Unpacking libjcode-pm-perl (2.06-1.1build1) ...
Selecting previously unselected package libmujs3:amd64.
Preparing to unpack .../57-libmujs3_1.3.3-3build2_amd64.deb ...
Unpacking libmujs3:amd64 (1.3.3-3build2) ...
Selecting previously unselected package libole-storage-lite-perl.
Preparing to unpack .../58-libole-storage-lite-perl_0.22-1_all.deb ...

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Unpacking libole-storage-lite-perl (0.22-1) ...
Selecting previously unselected package libparse-recdescent-perl.
Preparing to unpack .../59-libparse-recdescent-perl_1.967015+dfsg-4_all.deb ...
Unpacking libparse-recdescent-perl (1.967015+dfsg-4) ...
Selecting previously unselected package libptexenc1:amd64.
Preparing to unpack .../60-libptexenc1_2023.20230311.66589-9build3_amd64.deb ...
Unpacking libptexenc1:amd64 (2023.20230311.66589-9build3) ...
Selecting previously unselected package librsvg2-common:amd64.
Preparing to unpack .../61-librsvg2-common_2.58.0+dfsg-1build1_amd64.deb ...
Unpacking librsvg2-common:amd64 (2.58.0+dfsg-1build1) ...
Selecting previously unselected package rubygems-integration.
Preparing to unpack .../62-rubygems-integration_1.18_all.deb ...
Unpacking rubygems-integration (1.18) ...
Selecting previously unselected package ruby3.2.
Preparing to unpack .../63-ruby3.2_3.2.3-1ubuntu0.24.04.8_amd64.deb ...
Unpacking ruby3.2 (3.2.3-1ubuntu0.24.04.8) ...
Selecting previously unselected package ruby-rubygems.
Preparing to unpack .../64-ruby-rubygems_3.4.20-1_all.deb ...
Unpacking ruby-rubygems (3.4.20-1) ...
Selecting previously unselected package ruby.
Preparing to unpack .../65-ruby_1%3a3.2~ubuntu1_amd64.deb ...
Unpacking ruby (1:3.2~ubuntu1) ...
Selecting previously unselected package rake.
Preparing to unpack .../66-rake_13.0.6-3_all.deb ...
Unpacking rake (13.0.6-3) ...
Selecting previously unselected package ruby-net-telnet.
Preparing to unpack .../67-ruby-net-telnet_0.2.0-1_all.deb ...
Unpacking ruby-net-telnet (0.2.0-1) ...
Selecting previously unselected package ruby-webrick.
Preparing to unpack .../68-ruby-webrick_1.8.1-1ubuntu0.2_all.deb ...
Unpacking ruby-webrick (1.8.1-1ubuntu0.2) ...
Selecting previously unselected package ruby-xmlrpc.
Preparing to unpack .../69-ruby-xmlrpc_0.3.2-2_all.deb ...
Unpacking ruby-xmlrpc (0.3.2-2) ...
Selecting previously unselected package ruby-sdbm:amd64.
Preparing to unpack .../70-ruby-sdbm_1.0.0-5build4_amd64.deb ...
Unpacking ruby-sdbm:amd64 (1.0.0-5build4) ...
Selecting previously unselected package libruby3.2:amd64.
Preparing to unpack .../71-libruby3.2_3.2.3-1ubuntu0.24.04.8_amd64.deb ...
Unpacking libruby3.2:amd64 (3.2.3-1ubuntu0.24.04.8) ...
Selecting previously unselected package libruby:amd64.
Preparing to unpack .../72-libruby_1%3a3.2~ubuntu1_amd64.deb ...
Unpacking libruby:amd64 (1:3.2~ubuntu1) ...
Selecting previously unselected package libspreadsheet-writeexcel-perl.
Preparing to unpack .../73-libspreadsheet-writeexcel-perl_2.40-4_all.deb ...
Unpacking libspreadsheet-writeexcel-perl (2.40-4) ...
Selecting previously unselected package libunicode-map-perl.
Preparing to unpack .../74-libunicode-map-perl_0.112-13build4_amd64.deb ...

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Unpacking libunicode-map-perl (0.112-13build4) ...
Selecting previously unselected package libspreadsheet-parseexcel-perl.
Preparing to unpack .../75-libspreadsheet-parseexcel-perl_0.6600-1_all.deb ...
Unpacking libspreadsheet-parseexcel-perl (0.6600-1) ...
Selecting previously unselected package libsyntax2:amd64.
Preparing to unpack .../76-libsyntax2_2023.20230311.66589-9build3_amd64.deb ...
Unpacking libsyntax2:amd64 (2023.20230311.66589-9build3) ...
Selecting previously unselected package libteckit0:amd64.
Preparing to unpack .../77-libteckit0_2.5.12+ds1-1_amd64.deb ...
Unpacking libteckit0:amd64 (2.5.12+ds1-1) ...
Selecting previously unselected package libtexlua53-5:amd64.
Preparing to unpack .../78-libtexlua53-5_2023.20230311.66589-9build3_amd64.deb
...
Unpacking libtexlua53-5:amd64 (2023.20230311.66589-9build3) ...
Selecting previously unselected package libzip-0-13t64:amd64.
Preparing to unpack .../79-libzip-0-13t64_0.13.72+dfsg.1-1.2build1_amd64.deb
...
Unpacking libzip-0-13t64:amd64 (0.13.72+dfsg.1-1.2build1) ...
Selecting previously unselected package tex-common.
Preparing to unpack .../80-tex-common_6.18_all.deb ...
Unpacking tex-common (6.18) ...
Selecting previously unselected package lmodern.
Preparing to unpack .../81-lmodern_2.005-1_all.deb ...
Unpacking lmodern (2.005-1) ...
Selecting previously unselected package mupdf-tools.
Preparing to unpack .../82-mupdf-tools_1.23.10+ds1-1build3_amd64.deb ...
Unpacking mupdf-tools (1.23.10+ds1-1build3) ...
Selecting previously unselected package preview-latex-style.
Preparing to unpack .../83-preview-latex-style_13.2-1_all.deb ...
Unpacking preview-latex-style (13.2-1) ...
Selecting previously unselected package t1utils.
Preparing to unpack .../84-t1utils_1.41-4build3_amd64.deb ...
Unpacking t1utils (1.41-4build3) ...
Selecting previously unselected package teckit.
Preparing to unpack .../85-teckit_2.5.12+ds1-1_amd64.deb ...
Unpacking teckit (2.5.12+ds1-1) ...
Selecting previously unselected package tex-gyre.
Preparing to unpack .../86-tex-gyre_20180621-6_all.deb ...
Unpacking tex-gyre (20180621-6) ...
Selecting previously unselected package texlive-binaries.
Preparing to unpack .../87-texlive-
binaries_2023.20230311.66589-9build3_amd64.deb ...
Unpacking texlive-binaries (2023.20230311.66589-9build3) ...
Selecting previously unselected package texlive-base.
Preparing to unpack .../88-texlive-base_2023.20240207-1_all.deb ...
Unpacking texlive-base (2023.20240207-1) ...
Selecting previously unselected package texlive-fonts-recommended.
Preparing to unpack .../89-texlive-fonts-recommended_2023.20240207-1_all.deb ...

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Unpacking texlive-fonts-recommended (2023.20240207-1) ...
Selecting previously unselected package texlive-latex-base.
Preparing to unpack .../90-texlive-latex-base_2023.20240207-1_all.deb ...
Unpacking texlive-latex-base (2023.20240207-1) ...
Selecting previously unselected package libfontbox-java.
Preparing to unpack .../91-libfontbox-java_1%3a1.8.16-5_all.deb ...
Unpacking libfontbox-java (1:1.8.16-5) ...
Selecting previously unselected package libpdfbox-java.
Preparing to unpack .../92-libpdfbox-java_1%3a1.8.16-5_all.deb ...
Unpacking libpdfbox-java (1:1.8.16-5) ...
Selecting previously unselected package texlive-latex-recommended.
Preparing to unpack .../93-texlive-latex-recommended_2023.20240207-1_all.deb ...
Unpacking texlive-latex-recommended (2023.20240207-1) ...
Selecting previously unselected package texlive-pictures.
Preparing to unpack .../94-texlive-pictures_2023.20240207-1_all.deb ...
Unpacking texlive-pictures (2023.20240207-1) ...
Selecting previously unselected package texlive-latex-extra.
Preparing to unpack .../95-texlive-latex-extra_2023.20240207-1_all.deb ...
Unpacking texlive-latex-extra (2023.20240207-1) ...
Selecting previously unselected package texlive-plain-generic.
Preparing to unpack .../96-texlive-plain-generic_2023.20240207-1_all.deb ...
Unpacking texlive-plain-generic (2023.20240207-1) ...
Selecting previously unselected package tipa.
Preparing to unpack .../97-tipa_2%3a1.3-21_all.deb ...
Unpacking tipa (2:1.3-21) ...
Selecting previously unselected package texlive-xetex.
Preparing to unpack .../98-texlive-xetex_2023.20240207-1_all.deb ...
Unpacking texlive-xetex (2023.20240207-1) ...
Setting up libjcode-pm-perl (2.06-1.1build1) ...
Setting up session-migration (0.3.9build1) ...
Created symlink /etc/systemd/user/graphical-session-pre.target.wants/session-
migration.service → /usr/lib/systemd/user/session-migration.service.
Setting up fonts-lato (2.015-1) ...
Setting up fonts-noto-mono (20201225-2) ...
Setting up openjdk-21-jre-headless:amd64 (21.0.12.1+1-1~24.04.4) ...
Setting up fonts-texgyre-math (20180621-6) ...
Setting up libwoff1:amd64 (1.0.2-2build1) ...
Setting up libmujs3:amd64 (1.3.3-3build2) ...
Setting up libio-stringy-perl (2.111-3) ...
Setting up libxtst6:amd64 (2:1.2.3-1.1build1) ...
Setting up libijs-0.35:amd64 (0.35-15.1build1) ...
Setting up libgs-common (10.02.1-dfsg1-0ubuntu7.9) ...
Setting up libfontbox-java (1:1.8.16-5) ...
Setting up libxxf86dga1:amd64 (2:1.1.5-1build1) ...
Setting up rubygems-integration (1.18) ...
Setting up poppler-data (0.4.12-1) ...
Setting up libdigest-perl-md5-perl (1.9-5) ...
Setting up tex-common (6.18) ...

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update-language: texlive-base not installed and configured, doing nothing!
Setting up libunicode-map-perl (0.112-13build4) ...
Setting up libzip-0-13t64:amd64 (0.13.72+dfsg.1-1.2build1) ...
Setting up libjbig2dec0:amd64 (0.20-1ubuntu0.24.04.1) ...
Setting up libgumbo2:amd64 (0.12.0+dfsg-2build1) ...
Setting up libteckit0:amd64 (2.5.12+ds1-1) ...
Setting up libapache-pom-java (29-2) ...
Setting up libatspi2.0-0t64:amd64 (2.52.0-1build1) ...
Setting up ruby-net-telnet (0.2.0-1) ...
Setting up xfonts-encodings (1:1.0.5-0ubuntu2) ...
Setting up t1utils (1.41-4build3) ...
Setting up libtexlua53-5:amd64 (2023.20230311.66589-9build3) ...
Setting up libcrypt-rc4-perl (2.02-5) ...
Setting up fonts-dejavu-mono (2.37-8) ...
Setting up libidn12:amd64 (1.42-1ubuntu0.1) ...
Setting up fonts-dejavu-core (2.37-8) ...
Setting up fonts-texgyre (20180621-6) ...
Setting up libsvg2-common:amd64 (2.58.0+dfsg-1build1) ...
Setting up libkpathsea6:amd64 (2023.20230311.66589-9build3) ...
Setting up ruby-webrick (1.8.1-1ubuntu0.2) ...
Setting up libole-storage-lite-perl (0.22-1) ...
Setting up fonts-dejavu-extra (2.37-8) ...
Setting up libparse-recdescent-perl (1.967015+dfsg-4) ...
Setting up at-spi2-common (2.52.0-1build1) ...
Setting up fonts-lmodern (2.005-1) ...
Setting up libcarp-clan-perl (6.08-2) ...
Setting up fonts-droid-fallback (1:6.0.1r16-1.1build1) ...
Setting up libxcomposite1:amd64 (1:0.4.5-1build3) ...
Setting up ruby-xmlrpc (0.3.2-2) ...
Setting up libdate-manip-perl (6.95-1) ...
Setting up libgtk-3-common (3.24.41-4ubuntu1.3) ...
Setting up libsyntax2:amd64 (2023.20230311.66589-9build3) ...
Setting up libpotrace0:amd64 (1.16-2build1) ...
Setting up teckit (2.5.12+ds1-1) ...
Setting up gsettings-desktop-schemas (46.1-0ubuntu1) ...
Setting up libatk1.0-0t64:amd64 (2.52.0-1build1) ...
Setting up libpdfbox-java (1:1.8.16-5) ...
Setting up preview-latex-style (13.2-1) ...
Setting up libcommons-parent-java (56-1) ...
Setting up libcommons-logging-java (1.3.0-1ubuntu1) ...
Setting up xfonts-utils (1:7.7+6build3) ...
Setting up mupdf-tools (1.23.10+ds1-1build3) ...
Setting up libspreadsheet-writeexcel-perl (2.40-4) ...
Setting up libbit-vector-perl:amd64 (7.4-3build1) ...
Setting up libptexenc1:amd64 (2023.20230311.66589-9build3) ...
Setting up libspreadsheet-parseexcel-perl (0.6600-1) ...
Setting up texlive-binaries (2023.20230311.66589-9build3) ...
update-alternatives: using /usr/bin/xdvi-xaw to provide /usr/bin/xdvi.bin

```

```

(xdvi.bin) in auto mode
update-alternatives: using /usr/bin/bibtex.original to provide /usr/bin/bibtex
(bibtex) in auto mode
Setting up lmodern (2.005-1) ...
Setting up x11-utils (7.7+6build2) ...
Setting up libatk-wrapper-java (0.40.0-3build2) ...
Setting up texlive-base (2023.20240207-1) ...
mktexlsr: Updating /var/lib/texmf/ls-R-TEXLIVEDIST...
mktexlsr: Updating /var/lib/texmf/ls-R-TEXMFMAIN...
mktexlsr: Updating /var/lib/texmf/ls-R...
mktexlsr: Done.
tl-paper: setting paper size for dvips to a4:
/var/lib/texmf/dvips/config/config-paper.ps
tl-paper: setting paper size for dvipdfmx to a4:
/var/lib/texmf/dvipdfmx/dvipdfmx-paper.cfg
tl-paper: setting paper size for xdvi to a4: /var/lib/texmf/xdvi/XDvi-paper
tl-paper: setting paper size for pdftex to a4: /var/lib/texmf/tex/generic/tex-
ini-files/pdftexconfig.tex
Setting up tex-gyre (20180621-6) ...
Setting up libatk-bridge2.0-0t64:amd64 (2.52.0-1build1) ...
Setting up libdate-calc-perl (6.4-2) ...
Setting up libdate-calc-xs-perl:amd64 (6.4-2build4) ...
Setting up texlive-plain-generic (2023.20240207-1) ...
Setting up fonts-urw-base35 (20200910-8) ...
Setting up libatk-wrapper-java-jni:amd64 (0.40.0-3build2) ...
Setting up texlive-latex-base (2023.20240207-1) ...
Setting up texlive-latex-recommended (2023.20240207-1) ...
Setting up texlive-pictures (2023.20240207-1) ...
Setting up texlive-fonts-recommended (2023.20240207-1) ...
Setting up tipa (2:1.3-21) ...
Setting up libgs10-common (10.02.1~dfsg1-0ubuntu7.9) ...
Setting up texlive-latex-extra (2023.20240207-1) ...
Setting up texlive-xetex (2023.20240207-1) ...
Setting up libgs10:amd64 (10.02.1~dfsg1-0ubuntu7.9) ...
Setting up dvisvgm (3.2.1+ds-1build1) ...
Setting up rake (13.0.6-3) ...
Setting up libruby:amd64 (1:3.2~ubuntu1) ...
Setting up ruby-sdbm:amd64 (1.0.0-5build4) ...
Setting up libruby3.2:amd64 (3.2.3-1ubuntu0.24.04.8) ...
Setting up ruby3.2 (3.2.3-1ubuntu0.24.04.8) ...
Setting up ruby (1:3.2~ubuntu1) ...
Setting up ruby-rubygems (3.4.20-1) ...
Processing triggers for man-db (2.12.0-4build2) ...
Processing triggers for libgl1.0-0t64:amd64 (2.80.0-6ubuntu3.8) ...
Setting up libgtk-3-0t64:amd64 (3.24.41-4ubuntu1.3) ...
Setting up at-spi2-core (2.52.0-1build1) ...
Processing triggers for libgdk-pixbuf2.0-0:amd64 (2.42.10+dfsg-3ubuntu3.3) ...
Processing triggers for fontconfig (2.15.0-1.1ubuntu2) ...

```

```

Processing triggers for hicolor-icon-theme (0.17-2) ...
Processing triggers for libc-bin (2.39-0ubuntu8.9) ...
/sbin/ldconfig.real: /usr/local/lib/libtcm.so.1 is not a symbolic link

/sbin/ldconfig.real: /usr/local/lib/libtbbmalloc.so.2 is not a symbolic link

/sbin/ldconfig.real: /usr/local/lib/libtbbbind_2_0.so.3 is not a symbolic link

/sbin/ldconfig.real: /usr/local/lib/libur_loader.so.0 is not a symbolic link

/sbin/ldconfig.real: /usr/local/lib/libtbbbind.so.3 is not a symbolic link

/sbin/ldconfig.real: /usr/local/lib/libur_adapter_opencl.so.0 is not a symbolic
link

/sbin/ldconfig.real: /usr/local/lib/libumf.so.1 is not a symbolic link

/sbin/ldconfig.real: /usr/local/lib/libtcm_debug.so.1 is not a symbolic link

/sbin/ldconfig.real: /usr/local/lib/libhwloc.so.15 is not a symbolic link

/sbin/ldconfig.real: /usr/local/lib/libtbbmalloc_proxy.so.2 is not a symbolic
link

/sbin/ldconfig.real: /usr/local/lib/libur_adapter_level_zero.so.0 is not a
symbolic link

/sbin/ldconfig.real: /usr/local/lib/libtbb.so.12 is not a symbolic link

/sbin/ldconfig.real: /usr/local/lib/libur_adapter_level_zero_v2.so.0 is not a
symbolic link

/sbin/ldconfig.real: /usr/local/lib/libtbbbind_2_5.so.3 is not a symbolic link

Setting up libgtk-3-bin (3.24.41-4ubuntu1.3) ...
Processing triggers for ca-certificates-java (20240118) ...
[0.001s][warning][os,container] Cgroup memory controller path at
'/sys/fs/cgroup' seems to have moved to '/../jupyter-children', detected
limits won't be accurate
[0.001s][warning][os,container] Cgroup cpu controller path at '/sys/fs/cgroup'
seems to have moved to '/../jupyter-children', detected limits won't be
accurate
done.
Setting up openjdk-21-jre:amd64 (21.0.12.1+1-1~24.04.4) ...
Setting up openjdk-21-jdk-headless:amd64 (21.0.12.1+1-1~24.04.4) ...
Setting up default-jre-headless (2:1.21-75+exp1) ...
Setting up default-jre (2:1.21-75+exp1) ...
Processing triggers for tex-common (6.18) ...

```

Running updmap-sys. This may take some time... done.
Running mktexlsr /var/lib/texmf ... done.
Building format(s) --all.
This may take some time... done.

```
[19]: import os
import shutil

# Buscar en la subcarpeta 'notebooks'
subfolder = './notebooks'
if os.path.exists(subfolder):
    notebooks = [f for f in os.listdir(subfolder) if f.endswith('.ipynb')]
    if notebooks:
        nb_name = notebooks[0]
        path_to_nb = os.path.join(subfolder, nb_name)
        print(f"Encontrado el cuaderno: {path_to_nb}")
        print("Procediendo a la conversión a PDF...")

        # Convertir usando jupyter nbconvert
        !jupyter nbconvert --to pdf "{path_to_nb}"

        # Mover el PDF resultante a la carpeta actual para descargarlo
        ↪ fácilmente
        pdf_name = nb_name.replace('.ipynb', '.pdf')
        pdf_path_origin = os.path.join(subfolder, pdf_name)
        if os.path.exists(pdf_path_origin):
            shutil.move(pdf_path_origin, f"./{pdf_name}")
            print(f"\n¡Éxito! Tu archivo PDF se ha generado y guardado como:
            ↪ {pdf_name} en la carpeta actual.")
        else:
            print("\nSe ejecutó la conversión pero no se generó el archivo
            ↪ esperado en la ruta de origen.")
        else:
            print("No se encontró ningún archivo .ipynb dentro de la carpeta
            ↪ 'notebooks'.")
    else:
        print("La subcarpeta 'notebooks' no existe en el directorio actual.")
```

No se encontró ningún archivo .ipynb en esta carpeta.
Directorio actual: /content/drive/Othercomputers/Mi portátil/cuadernos_extra
Contenido del directorio: ['.git', 'notebooks', 'images', '.gitignore']